

# POLITECNICO DI TORINO

Master's Degree in Electronics



Master's Degree Thesis

## Loss analysis of a 4-level, flying capacitor, bridge-less, totem-pole PFC

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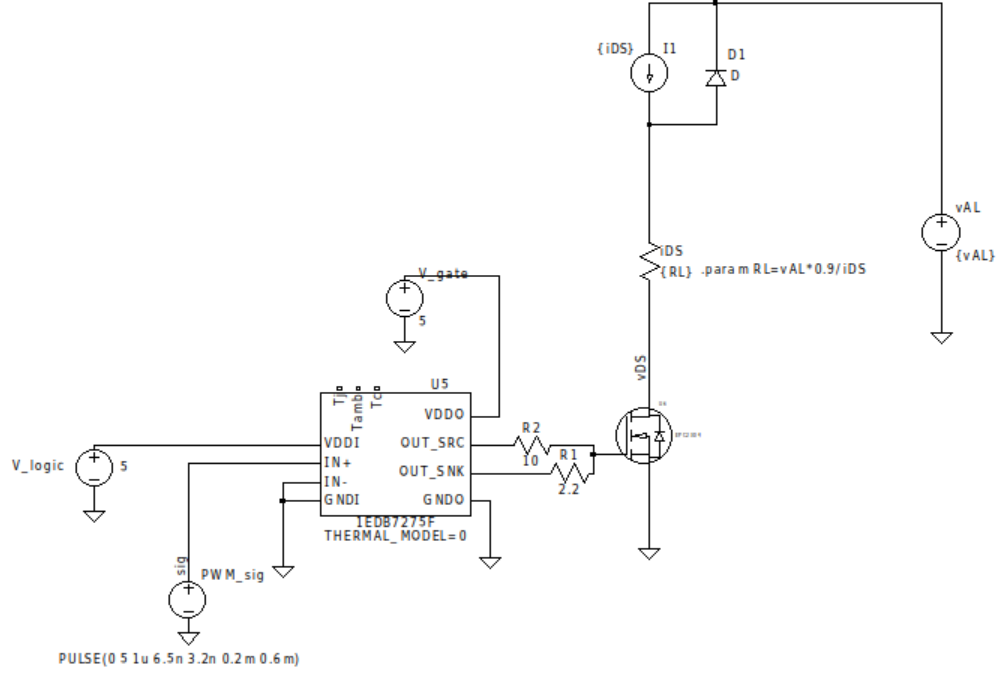


# Table of Contents

|          |                                             |           |
|----------|---------------------------------------------|-----------|
| <b>1</b> | <b>Simulations overview</b>                 | <b>3</b>  |
| 1.1      | Simulation setup . . . . .                  | 3         |
| 1.1.1    | Slow-Leg MOSFET SPICE simulations . . . . . | 3         |
| 1.1.2    | Fast-Leg MOSFET SPICE simulations . . . . . | 4         |
| 1.1.3    | EMI filter considerations . . . . .         | 4         |
| 1.1.4    | PFC simulation . . . . .                    | 6         |
| 1.2      | Simulation results . . . . .                | 7         |
|          | <b>List of Tables</b>                       | <b>11</b> |
|          | <b>List of Figures</b>                      | <b>12</b> |



### 1.1.2 Fast-Leg MOSFET SPICE simulations

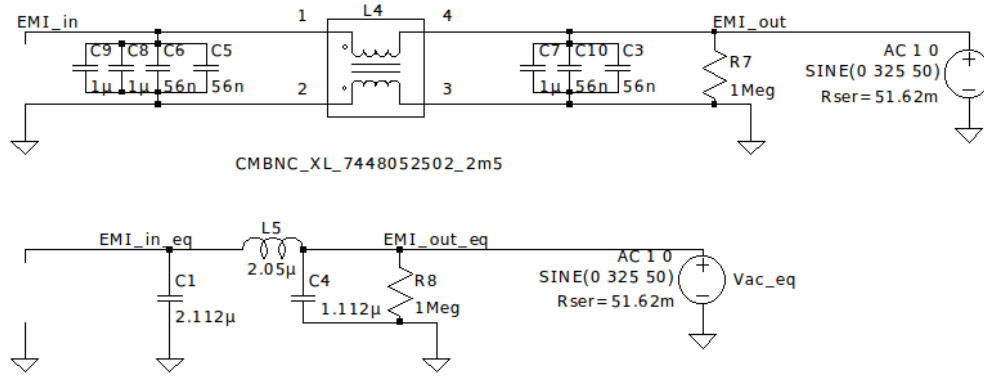


**Figure 1.2:** SPICE simulation for GaN Energy

As previously done for the slow leg Silicon MOSFETs , also the GaN-FETs have been simulated using the same technique, doing so all the passive/filtering elements, from the controller, to the actual FETs, are summarized in a single number which are the conduction/switching losses , which are then used by PLECS for the loss analysis.

### 1.1.3 EMI filter considerations

In order to improve computational time with the PLECS software , some modifications to the EMI filter have been performed. In the simulation enviroment no Protective Earth is given , and the Ground reference is designated for the output of the converter. Doing so any reasoning about common mode noise is futile, since all elements are referred to the same ground. Only differential noise have been studied, which is coherent with the 91107 efficiency and iTHD measures , which have been performed in an isolated enviroment where no ground connection was provided.



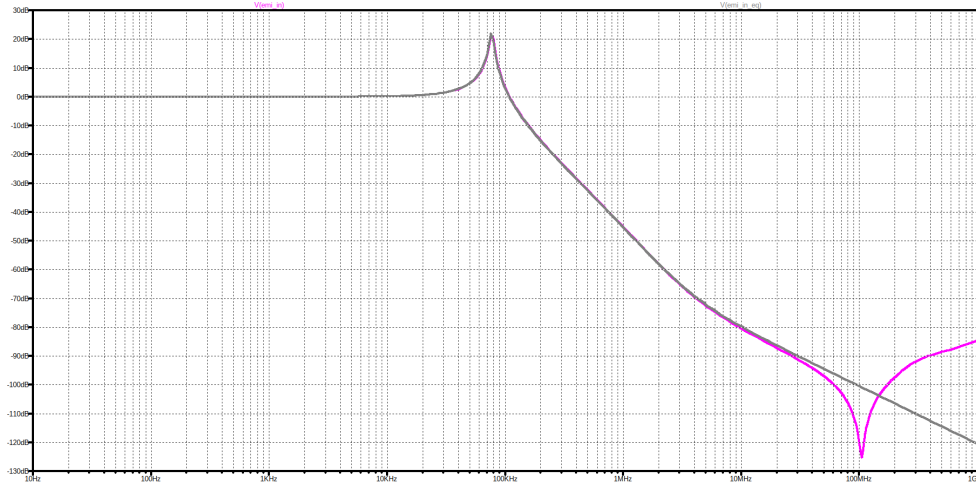
**Figure 1.3:** EMI filter equivalent

Since only differential mode is considered the Common mode choke can be simulated as a single inductance. The circuit shown here [1.3] is used to compare the filter with the complete SPICE model of all the components which compose the EMI filter , with the simplified version of the same. The most relevant quantities (such capacitor and inductor ESR) have been kept identical to the complete SPICE model.

The generator equivalent resistance has been calculated from the sum of all active and passive devices in series with the power path + 10% in order to account for PCB ,connectors and solder joint parasitic resistances.

The Output to input transfer function has been calculated in order to visualize the filtering behaviour.

The relevant metric is the attenuation factor at 420kHz which is the inductor ripple frequency.



**Figure 1.4:** EMI Out to In transfer function

The input impedance is also computed[1.5].  
The equivalent generator DOES NOT have an ESR in order to further improve PLECS simulation

time, the "ideal" model of the grid generator does have a  $50m\Omega$  value in order to account parasitics in the grid connection.

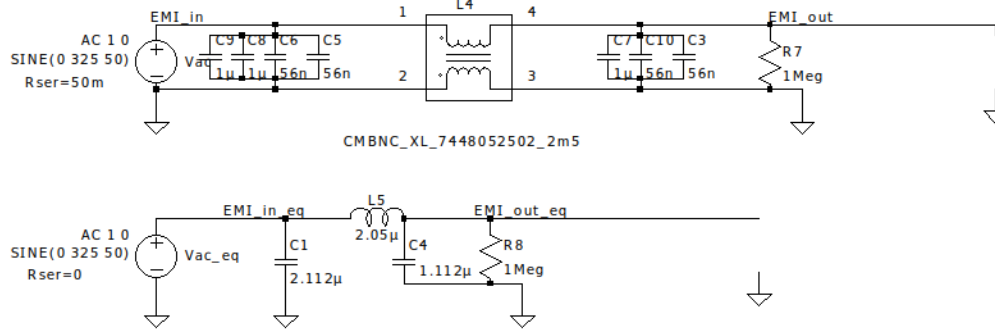


Figure 1.5: EMI input impedance circuit

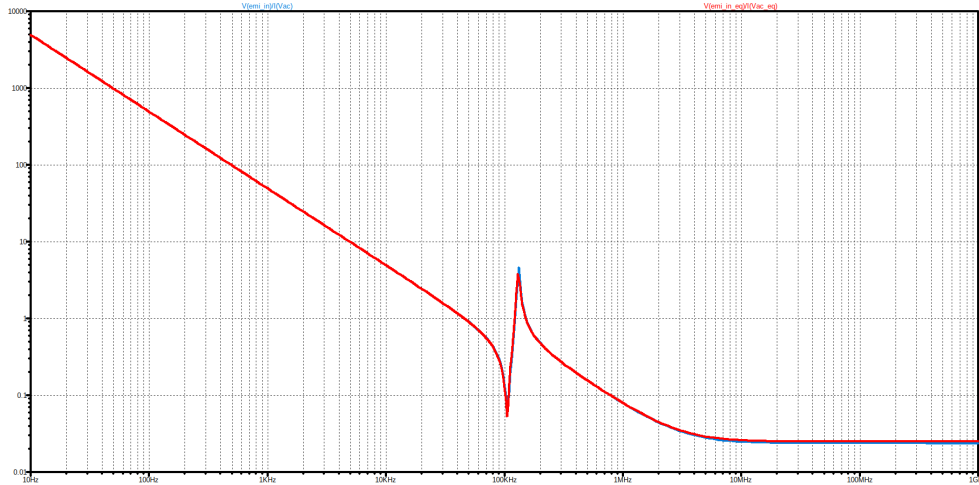


Figure 1.6: EMI input impedance

### 1.1.4 PFC simulation

The entire PFC system was simulated using the PLECS software, this is much faster than any SPICE simulation, since PLECS runs on a system level, and the simulated devices are either on or off, no transients are present, that's why the energy-on/energy-off SPICE simulations were fundamental, since those data are needed by the PLECS software to perform any kind of loss analysis.

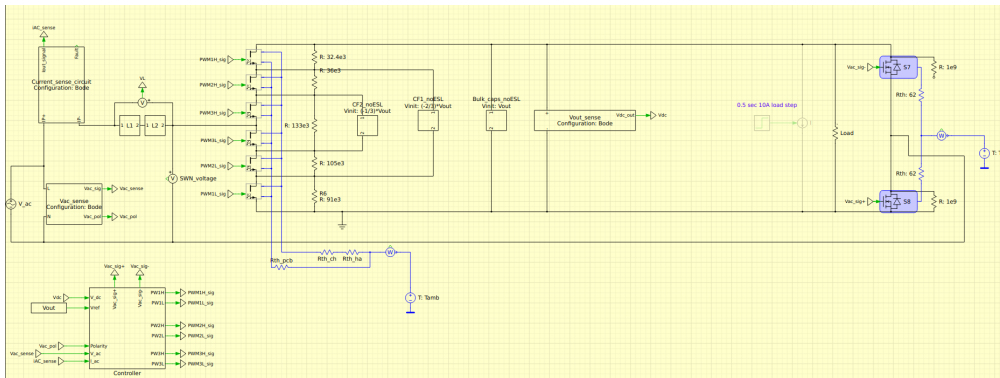


Figure 1.7: PFC PLECS schematic

## 1.2 Simulation results

These simulations DO NOT take into account:

- House-keeping PSU stand-by power and efficiency
- Gate driver consumption
- Sensing circuits
- Controller consumption

Only the power section components have been included with some limitations:

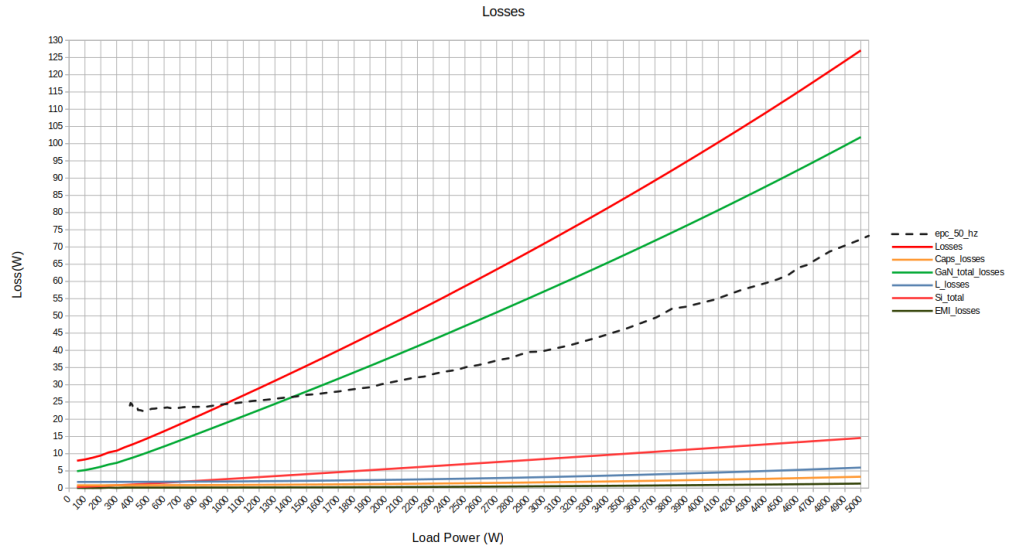
- NO ESL on Flying caps
- NO ESL on HF decoupling caps
- NO ESL on Electrolytic bulk caps
- NO AC losses models for inductors



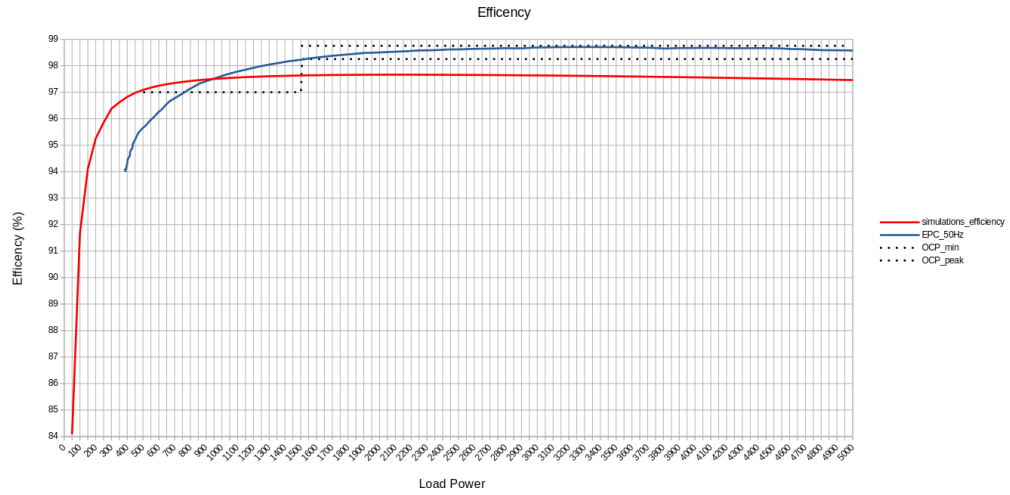
All remaining elements have been modeled as closely as possible to the EPC91107 datasheet

- feedback filtering chain (low pass filters as transfer functions)
- controller to gate driver filtering chain(included in GaN/Si energy calculations)
- devices parasitic

A swept power analysis have been performed from 50W to 5000W.



**Figure 1.8:** Losses summary



**Figure 1.9:** Efficiency in %

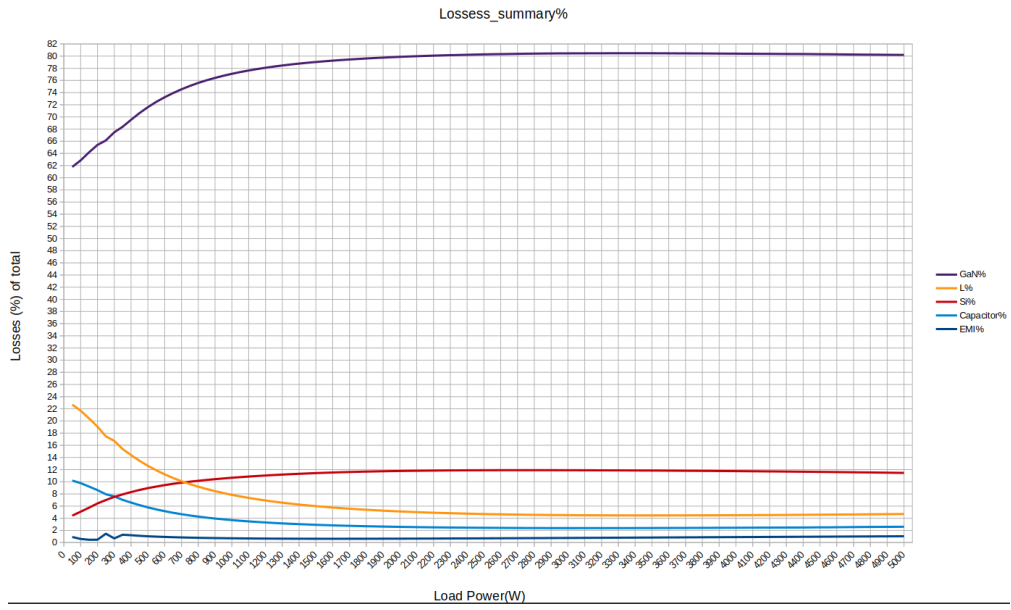


Figure 1.10: Losses summary % of total

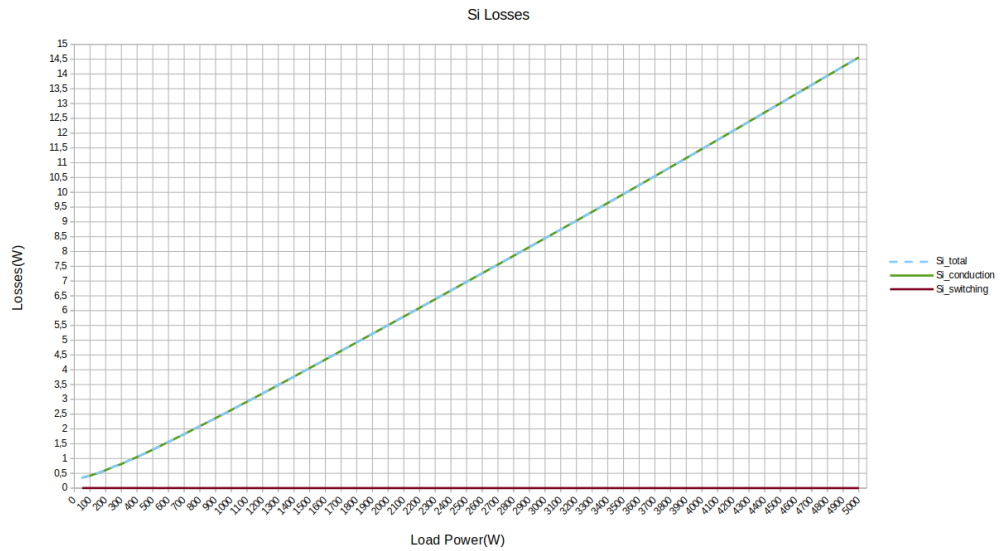


Figure 1.11: Si losses detail



Figure 1.12: GaN losses detail

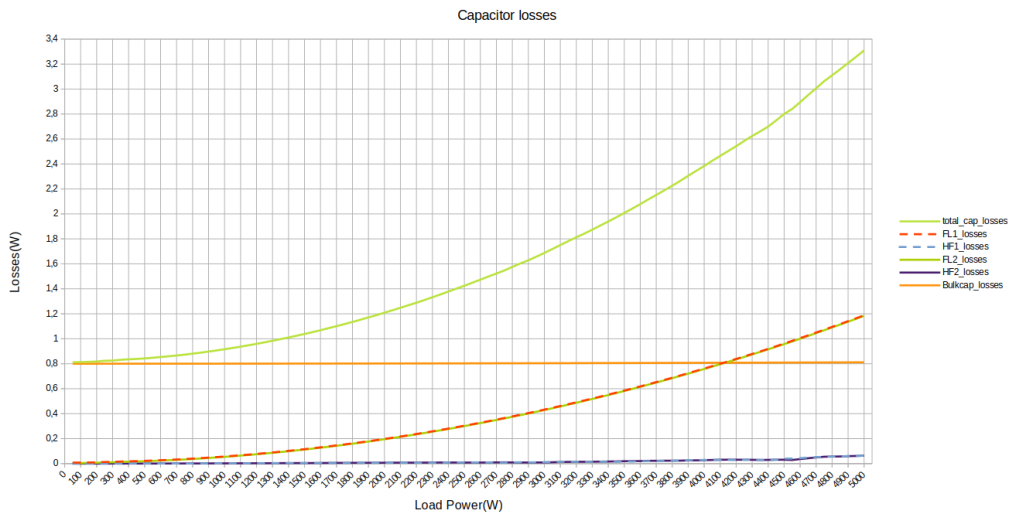


Figure 1.13: Capacitors losses detail

# List of Tables

# List of Figures

|      |                                           |    |
|------|-------------------------------------------|----|
| 1.1  | SPICE simulation for Si Energy . . . . .  | 3  |
| 1.2  | SPICE simulation for GaN Energy . . . . . | 4  |
| 1.3  | EMI filter equivalent . . . . .           | 5  |
| 1.4  | EMI Out to In transfer function . . . . . | 5  |
| 1.5  | EMI input impedance circuit . . . . .     | 6  |
| 1.6  | EMI input impedance . . . . .             | 6  |
| 1.7  | PFC PLECS schematic . . . . .             | 7  |
| 1.8  | Losses summary . . . . .                  | 8  |
| 1.9  | Efficiency in % . . . . .                 | 8  |
| 1.10 | Losses summary % of total . . . . .       | 9  |
| 1.11 | Si losses detail . . . . .                | 9  |
| 1.12 | GaN losses detail . . . . .               | 10 |
| 1.13 | Capacitors losses detail . . . . .        | 10 |